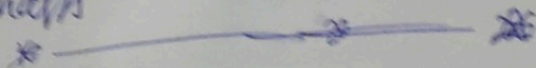


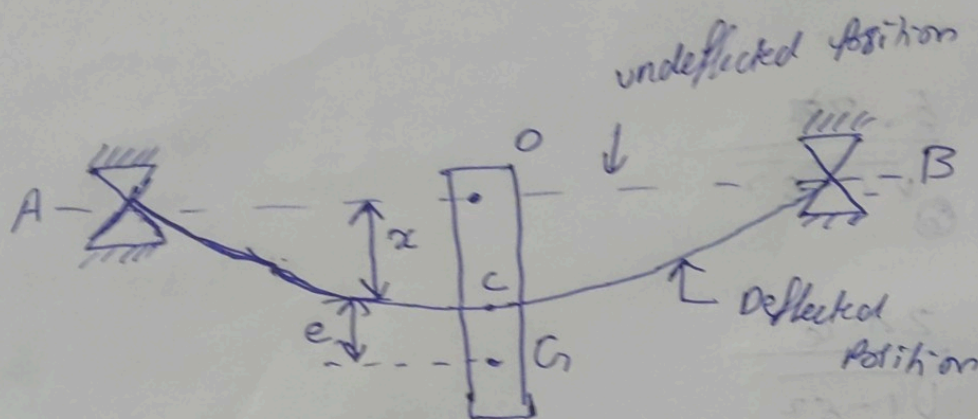
MODULE - 3

Transverse Vibration of Shaft & Whirling Speed of

Shafts



Critical Speed of Light Shaft having single disc
without damping



$$Kx = \frac{W}{g} (\omega^2) (x + e)$$

$$Kx = \frac{W}{g} \omega^2 x = \frac{W}{g} \omega^2 e$$

$$\frac{x}{e} = \frac{\frac{W}{g} \cdot \omega^2}{K - \frac{W}{g} \omega^2} = \frac{1}{\frac{Kg}{W \cdot \omega^2} - 1}$$

$$\frac{x}{e} = \frac{1}{\left(\frac{\omega_c}{\omega}\right)^2 - 1}$$

$$\text{Assume } \omega_c^2 = \frac{Kg}{W} = \frac{g}{\delta}$$

$\delta \rightarrow$ Static deflection

$$\boxed{\omega_c = \sqrt{g/\delta}} \rightarrow \text{whirling or critical speed.}$$

$$\boxed{\delta = \frac{\left(W + \frac{17}{35} P l g \right) l^3}{48 EI}}$$

$\delta = \text{mass/unit length}$

$$I \rightarrow \frac{\pi \cdot d^4}{64}$$

$$k = \frac{48 EI}{l^3} \text{ for shaft}$$

Critical Speed

When the ~~critical~~ natural frequency of system coincides with external force frequency, it is called resonance. The speed at which resonance occurs is known as critical speed.

at resonance, $\omega = \omega_n$

$$\text{damping factor, } \xi = \frac{c}{2m\omega_n}$$

$$\tan \phi = \alpha, \quad \phi = \pi/2$$

$$\text{Amplitude at resonance, } = \frac{F}{C\omega_n}$$

$\Rightarrow$ damping Co-efficient

at critical speed, the amplitude of vibration of rotor is excessively large and large amount

of amplitude is transmitted to the foundation. It may occur because of eccentric mounting of rotor, non uniform distribution of rotor material, bending of shaft etc.

- 1) A machine weighing 3.5 kg vibrates in a viscous medium. A harmonic exciting force of 40 N acts on the machine and produces a resonant amplitude of 18 mm; with a period of 0.25. Determine the damping coefficient.

Given $m = 3.5 \text{ kg}$

$F = 40 \text{ N}$

resonant amplitude, $x_r = 18 \times 10^{-3} \text{ m}$.

Time Period $T = 0.25$

~~damping factor, $\delta = \frac{c}{2m\omega_n}$~~

$$x_r = \frac{F}{C\omega_n}$$

$$18 \times 10^{-3} = \frac{40}{C \times \omega_n}$$

$$\omega_n = \sqrt{k/m}$$

$$\omega_n = 2\pi f = \frac{2\pi}{T} = \frac{2\pi}{0.25} = 31.41 \text{ rad/s}$$

$$f = \frac{1}{T} \text{ (frequency is reciprocal of time period)}$$

$$1.8 \times 10^{-3} = \frac{L_0}{C \times 31.41}$$

$$C = 70.74 \text{ rpm/m}$$

- 2) A shaft of 2.5 cm diameter, freely supported by bearings 75 cm apart, carries a single concentrated load of 196.2 N, midway between the bearings. Determine the first critical speed, assume that shaft material has a density of $8 \times 10^3 \text{ kg/m}^3$ and G is $2.1 \times 10^{11} \text{ N/m}^2$.

Ans-1) Critical speed, $\omega_c = \sqrt{g/l}$

$$f = \frac{\left(W + \frac{17}{35} \cdot P l g \right) \times l^3}{48 \cdot E I}$$

$$W = 196.2 \text{ N}$$

$$l = 0.75 \text{ m}$$

$$g = 9.81 \text{ m/s}^2$$

$$E = 2.1 \times 10^{11} \text{ N/m}^2$$

$P = 8 \times 10^3 \text{ kg/m}^3$, convert to kg/m by multiplying with area.

$$\therefore P = 8 \times 10^3 \times \pi r^2$$

$$= 8 \times 10^3 \times \pi \times (1.25 \times 10^{-2})^2$$

$$= 3.926 \text{ kg/m}$$

$$\omega_c = \sqrt{\frac{g}{l}}$$

$$I = \frac{\pi \times d^4}{64} = \frac{\pi \times (2.5 \times 10^{-2})^4}{64} = 1.916 \times 10^{-8} \text{ m}^4$$

$$\therefore \delta = \frac{(196.2 + \frac{17}{25} \times 3.926 \times 10^{-7} \times 9.81)}{4.8 \times 2.1 \times 10^{11} \times 1.916 \times 10^{-8}}$$

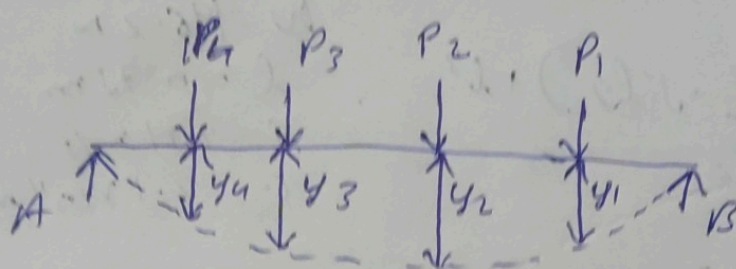
$$= 4.592 \times 10^{-4} \text{ m}$$

$$\omega_c = \sqrt{\frac{9.81}{4.592 \times 10^{-4}}} = 146.16 \text{ rad/s}$$

$$\therefore f_c = \frac{\omega_c}{2\pi} = 23.27 \text{ Hz}$$

*1/ Approximate method to analyse vibration system

a) Rayleigh's method



This is the energy method to find frequency. This method is used to find natural frequency of the system when transverse point loads are acting on the beam or shaft.

The max Potential energy :

$$[P = mg]$$

$$PE = \frac{1}{2} P_1 y_1 + \frac{1}{2} P_2 y_2 + \frac{1}{2} P_3 y_3 + \dots$$

$$= \frac{1}{2} \sum P y$$

$$KE = \frac{1}{2} \frac{P_1}{g} (\omega_n y_1)^2 + \frac{1}{2} \frac{P_2}{g} (\omega_n y_2)^2 + \dots$$

$$= \frac{\omega_n^2}{2g} \sum P y^2$$

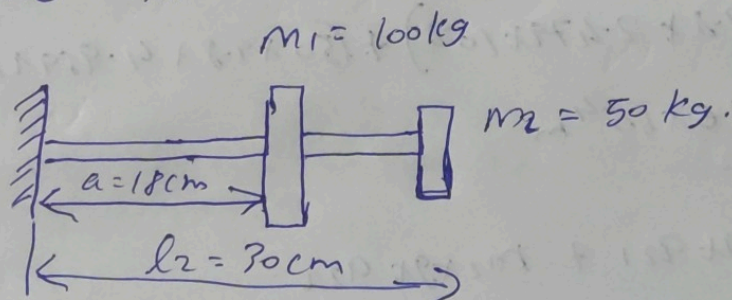
on equating,

$$\omega_n = \sqrt{\frac{g \sum P y}{\sum P y^2}}$$

1) Find the lowest natural frequency of vibration for the system shown in figure 6-9 by Rayleigh's method.

$$E = 1.96 \times 10^{11} \text{ N/m}^2$$

$$I = 4 \times 10^{-7} \text{ m}^4$$



$$\omega_n = \sqrt{\frac{g \sum P y}{\sum P y^2}}$$

ii,

$$\omega_n = \sqrt{\frac{g \sum P_i y_i}{\sum (P_i y_i^2 + P_j y_j^2)}}$$

$$y_1 = m_1 \cdot g \cdot a_{11} + m_2 g \cdot a_{12} \quad \& \quad y_2 = m_1 \cdot g \cdot a_{21} + m_2 g \cdot a_{22}$$

$$m_1 = 100 \text{ kg}, \quad m_2 = 50 \text{ kg}, \quad g = 9.8$$

$$a_{11} = \frac{a^3}{3EI} = \frac{(18 \times 10^{-2})^3}{3 \times 1.96 \times 10^{11} \times 4 \times 10^{-7}} = 2.479 \times 10^{-8}$$

$$a_{12} = a_{21} = \frac{a^2 \times (3l - a)}{6EI} = \frac{(18 \times 10^{-2})^2 \times (3 \times 0.30 - 0.18)}{6 \times 1.96 \times 10^{11} \times 4 \times 10^{-7}} \\ = 4.959 \times 10^{-8}$$

$$a_{22} = \frac{l^3}{3EI} = \frac{(0.30)^3}{3 \times 1.96 \times 10^{11} \times 4 \times 10^{-7}} = 1.147 \times 10^{-7}$$

$$y_1 = m_1 g x a_{11} + m_2 g x a_{12}$$

$$= (100 \times 9.8 \times 2.479 \times 10^{-8}) + (50 \times 9.8 \times 4.959 \times 10^{-8}) \\ = 0.486 \times 10^{-4} \text{ m}$$

$$y_2 = m_1 g x a_{21} + m_2 g x a_{22}$$

$$= (100 \times 9.8 \times 4.959 \times 10^{-8}) + (50 \times 9.8 \times 1.147 \times 10^{-7}) \\ = 10.48 \times 10^{-5} \text{ m}$$

$$\begin{aligned} p_1 &= m_1 x g \\ p_2 &= m_2 x g \end{aligned}$$

$$\therefore \omega = \sqrt{\frac{9.8 (100 \times 9.8 \times 0.486 \times 10^{-4}) + 50 \times 9.8 \times 10.48 \times 10^{-5}}{(100 \times 9.8 \times (0.486 \times 10^{-4})^2 + 50 \times 9.8 \times (10.48 \times 10^{-5})^2)}} \\ = 355 \text{ rad/s}$$

DUNKERLEY'S METHOD

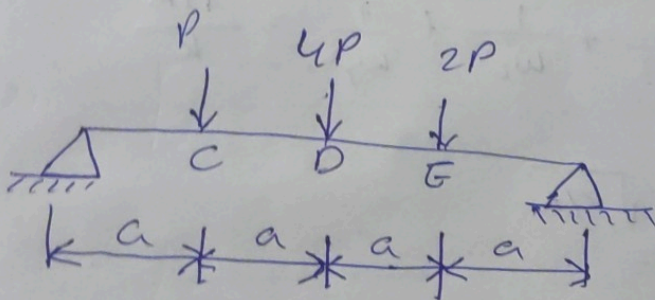
It is used to find out natural frequency of structure. The load of the system is uniformly distributed ~~mass~~ Dunkerley's eqn.

$$\frac{1}{\omega_n^2} = \frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \dots + \frac{1}{\omega_s^2}$$

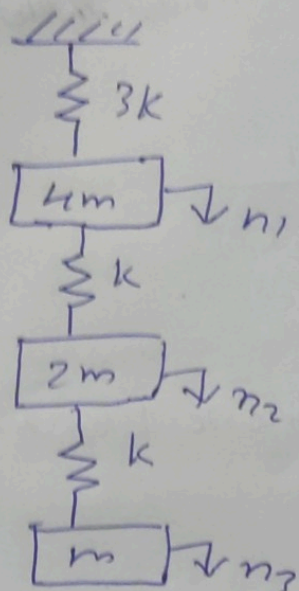
When ω_n = natural frequency of transverse vibration of shaft for many point loads.

$\omega_1, \omega_2, \omega_3 \dots$ = natural frequency of individual point loads.

ω_s = natural frequency of transverse vibration because of the weight of shaft.



Q) Use Dunkerley's formula to find the lowest natural frequency of the system shown below.



Ans)

$$\omega_0 = \sqrt{k/m}$$

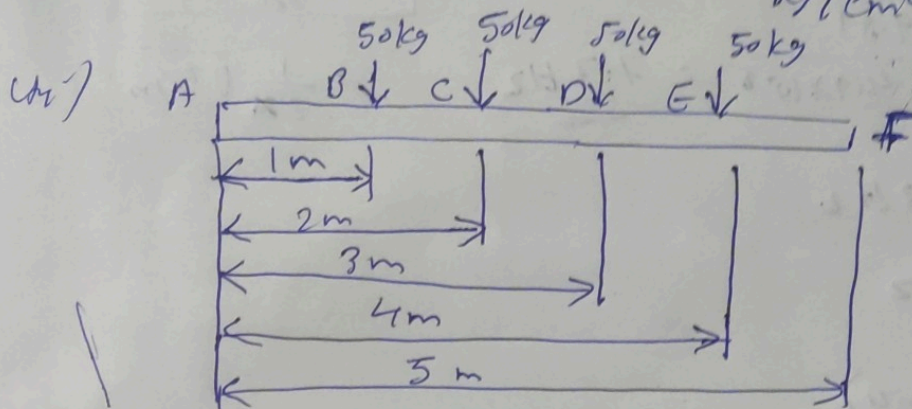
$$\omega_1 = \sqrt{3k/4m}, \quad \omega_2 = \sqrt{k/2m}, \quad \omega_3 = \sqrt{k/m}$$

It is given:

$$\therefore \frac{1}{\omega^2} = \frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \frac{1}{\omega_3^2} = \frac{13m}{k}$$

$$\therefore \omega = 0.475 \sqrt{k/m}$$

Q.) A shaft of negligible weight 6 cm dia and 5 m long is simply supported at the ends and carries four weights 50 kg each at equal distance over the length of the shaft find the frequency of vibration. Take $E = 2 \times 10^6 \text{ kg/cm}^2$.



$$\text{moment of inertia, } I = \frac{\pi}{64} \cdot d^4 = \frac{\pi}{64} \times 6^4 = \underline{63.585 \text{ cm}^4}$$

The general expression for static deflection because of point load w is given by

$$y = \frac{W \cdot l_1^2 \times l_2^2}{3 \cdot E \cdot I \cdot l}$$

$$y_B = \frac{50 \times 100^2 \times 400^2}{3 \times 2 \times 10^6 \times 63.585 \times 500} = \underline{0.419 \text{ cm}}$$

$$y_E = \frac{50 \times 200^2 \times 300^2}{3 \times 2 \times 10^6 \times 63.585 \times 500} = \underline{0.948 \text{ cm}}$$

$$y_D = 0.948 \text{ cm}$$

$$y_C = 0.419 \text{ cm}$$

General expression for natural frequency is given by:

$$\text{natural frequency, } \omega_n = \frac{1}{2\pi} \sqrt{g/y} \quad \text{r/s.}$$

$$\text{natural frequency } f_n = \frac{\omega}{2\pi} \text{ Hz}$$

$$f = \frac{1}{2\pi} \omega_n$$

$$\text{So, } f_0 = \frac{1}{2\pi} \sqrt{\frac{9.81}{0.419 \times 10^{-2}}} = \underline{7.7 \text{ Hz}} = \frac{1}{2\pi} \sqrt{k/m}$$

$$f_c = f_0 = 5.13 \text{ Hz}$$

$$f_c = 7.7 \text{ Hz}$$

For the system (series)

$$\frac{1}{f_n^2} = \frac{1}{f_1^2} + \frac{1}{f_2^2} + \frac{1}{f_3^2} + \dots$$

$$\frac{1}{f_n^2} = \frac{1}{(7.7)^2} + \frac{1}{(5.13)^2} + \frac{1}{(5.13)^2} + \frac{1}{(7.7)^2}$$

$$= 0.03377 + 0.7599$$

$$f_n^2 = 9.113; \quad \underline{f_n = 3.01 \text{ Hz}}$$